Deep Space Network Telecommunication CubeSat Antenna: Using the deployable Ka-band mesh reflector antenna.

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BACKGROUND IMAGE—©GRAPHIC STOCK

The Deep-Space Network Telecommunication CubeSat Antenna

Using the deployable Ka-band mesh reflector antenna.

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In the near future, CubeSats will be deployed beyond low-Earth orbit (LEO) to perform scientific tasks in deep space. To do so, these CubeSats will need high-gain antennas (HGAs) that fit in a highly confined volume. In this article, we present an innovative, deployable Ka-band antenna that folds in a 1.5-U ($10 \times 10 \times 15$ cm^3) stowage volume suitable for 6U ($10 \times 20 \times 30$ cm^3)-class CubeSats. This antenna is designed for telecommunication and is compatible with NASA's deep-space network (DSN) at Ka-band frequencies (i.e., uplink: 34.2–34.7 GHz; downlink: 31.8–32.3 GHz). Calculations and measurements show that 42.0-dBi gain and 57% aperture efficiency are obtained at 32 GHz.

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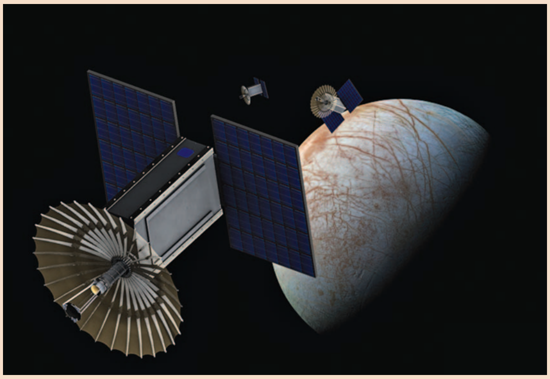
We thoroughly describe the mechanical deployment mechanism because it is a critical component of the deployable CubeSat antenna. This challenging new design evolved from our previous design that only provided linear polarization and a single-frequency band.

AN ANTENNA FOR LEO AND DEEP-SPACE HIGH-DATA-RATE COMMUNICATIONS

NASA is currently investigating the possibility of sending a fleet of CubeSats fanning out to the moon or other deep-space destinations such as Mars, Jupiter, or Europa. CubeSats are cost-effective solutions for space exploration that are capable of carrying out meaningful scientific experiments across the solar system. For instance, the first interplanetary CubeSat, Mars Cube One (MarCO), is planned for launch in 2018 alongside NASA's InSight Mars lander mission [1]. The goal of MarCO is to provide a communication link to Earth for the InSight mission during the entry descent and landing phase of the lander.

A telecommunication system compatible with the NASA DSN is required to venture beyond LEO. This system calls for HGAs that provide at least 28.0-dBi gain at X-band [1] or 42 dBi at Ka-band. In addition, these HGAs must stow in a highly constrained volume, typically <1.5 U, or $10 \times 10 \times 15$ cm^3. Most LEO CubeSat communication systems use dipole antennas at ultrahigh frequency or patch antennas at S-band that provide ~0–6-dBi gain.



(a)

(b)

FIGURE 1. (a) An artist's rendering of the constellation of 6-U CubeSats ($10 \times 20 \times 30$ cm^3) using the proposed deployable reflector antenna in orbit around Europa. (b) The antenna prototype.

As far as earth science applications are concerned, leveraging the recent progress in radio detection and ranging (radar) miniaturization, a LEO precipitation radar known as RainCube [2] is currently in development at NASA's Jet Propulsion Laboratory. Planned for a 2018 launch and expected to be the first radar flown in a CubeSat, RainCube will allow monitoring of the evolution of weather systems over a short time scale. To enable this technology, a deployable mesh reflector antenna has been designed at 35.75 GHz with linear polarization [2].

Three deployable antenna technologies have been widely investigated: deployable reflectarray [1], inflatable [3], and deployable mesh reflector [2], [4], [5]. Although an inflatable antenna is unlikely to satisfy the required surface accuracy at Ka-band, the deployable reflectarray and mesh reflector were proven to be successful solutions [1], [2]. Following the successful RainCube antenna development [2], we present here an updated version of the deployable mesh reflector for DSN telecommunication that requires a dual frequency band and right-hand circular polarization (RHCP). The performance of the telecommunication antenna should be maximized at downlink frequencies, where the signal-to-noise ratio is smaller. The new design presented in this article is accompanied by the following new features: 1) RHCP using a customized polarizer, 2) two frequency bands (uplink and downlink), and 3) more reliable mechanical deployment. The volume and weight limitations drive the electromagnetic and mechanical choices. This antenna will pave the way for high-data-rate communication for LEO and deep-space missions. As an example, for illustration purposes, Figure 1(a) shows a constellation of CubeSat in orbit around Europa.

ANTENNA DESIGN

ANTENNA CONFIGURATION

The fabricated antenna is shown in Figure 1(b). The reflector antenna is optimized at the Ka-band downlink (31.8–32.3 GHz) and uplink (34.2–34.7 GHz) frequency bands to minimize the following:

- spillover and taper loss
- subreflector, feed, and struts blockage loss
- subreflector diffraction effect
- surface aberration caused by the limited number of ribs (i.e., 30 ribs).

The stowed antenna assembly is designed to fit into a 1.5-U volume ($10 \times 10 \times 15$ cm^3) and deploy into a 0.5-m reflector antenna. To minimize the mechanical deployment difficulty, an axially symmetrical reflector is preferred. The design follows similar steps as was used in [2] for the linear polarized configuration; however, this antenna has more challenging performance requirements. To assist readers in following the steps taken to arrive at the optimized design, we elaborate on the various important design considerations.

The following two constraints were set by the mechanical deployment:

1) The F/D ratio, where F is the focal length and D the main reflector diameter, is selected to minimize the rib curvature. This is crucial to ensure that the ribs fit within the allocated volume between the folded horn and the cylindrical canister (Figure 2). We found that an F/D ratio of 0.5 can be accommodated and selected a 0.5-m-diameter paraboloidal main reflector with a focal length F_{ideal} of 0.25 m. The focal length is a tradeoff between best radio-frequency (RF) performance [i.e., to minimize the sidelobe level (SLL) and taper loss] and mechanical deployment.
2) The height of the subreflector is driven by the height of the stowed volume and the number of deployment mechanisms necessary to deploy the subreflector. To limit the feed deployment to only one deployment mechanism, the subreflector must be at a maximum of 22 cm above the reflector vertex. In addition, the distance between the vertex and the tip of the horn should stay within 12 cm, as shown in Figures 2 and 3. This can be effectively satisfied using a Cassegrain design, where the subreflector is below the focal point (see Figure 2). This places the subreflector at a distance of 0.22 m above the vertex.

The Ka-band deployable reflector antenna consists of the following six main elements (see Figure 2):

- multiflare RHCP feed horn
- three struts
- hyperbolic subreflector
- 0.5-m deployable mesh reflector
- telescoping waveguide
- polarizer.

The deployable mesh reflector is constructed using 30 ribs and, consequently, is not a true parabolic surface. The maximum number of ribs is limited by the available stowage volume. Although the ribs are perfectly parabolic, the mesh stretches between each rib, resulting in a flat surface. This has the effect of spreading the focal point of the parabolic reflector into a focal region [6]. Therefore, the subreflector shape needs to be reoptimized when using 30 ribs.

The subreflector is defined by a diameter d_{sub} of 73 mm, a vertex distance of 80 mm, and a foci distance of 138 mm. The subreflector was slightly extended to increase the antenna efficiency as shown in [7]. This leads to a gain increase of 0.1 dB. The subreflector diameter is about 0.146 times the reflector diameter. The overall dimensions of the optimized antenna design are shown in Figure 4. The maximum possible antenna gain can be expressed in terms of the physical area of the aperture by $G_{\max} = (\pi \cdot D/\lambda)^2$, where λ is the free-space wavelength. The maximum directivity of a 0.5-m diameter reflector is 44.5 and 45.1 dBi at 32 and 34.45 GHz, respectively.

OPTIMIZATION OF THE DEPLOYABLE MESH REFLECTOR

The feed is a multiflare horn, in which the telescoping waveguide can easily fit inside the horn when folded (see Figure 2). Multiflare horns provide good beam circularity, stable feed taper, and low cross polarization [8]. The optimized feed provides a minimum feed taper of –10 dB at 17° at 32 GHz to minimize taper and SLL (Figure 5). The circular polarization is obtained using the orthomode transducer (OMT) polarizer shown in Figure 6 [9], [10]. The OMT was optimized to operate at the DSN frequency bands. Although it can generate both RHCP and left-hand circular polarization (LHCP), only RHCP is required. Depending on the cross-polar discrimination mission requirement, the LHCP port can be either shorted or terminated with a

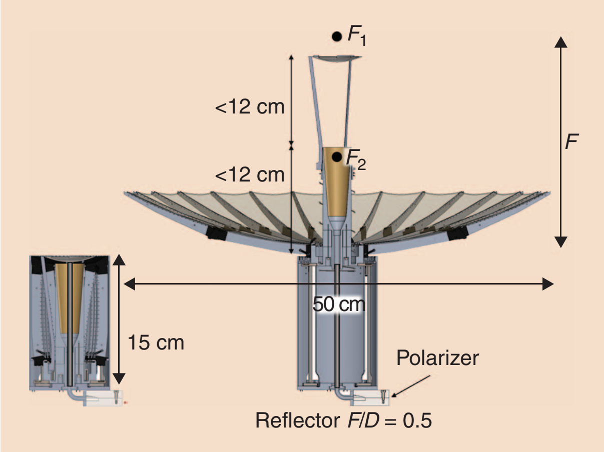


FIGURE 2. A stowage volume and constraint on the Cassegrain dimensions. F_1 and F_2 are the focal points of the subreflector. F_1 is also the focal point of the main reflector.

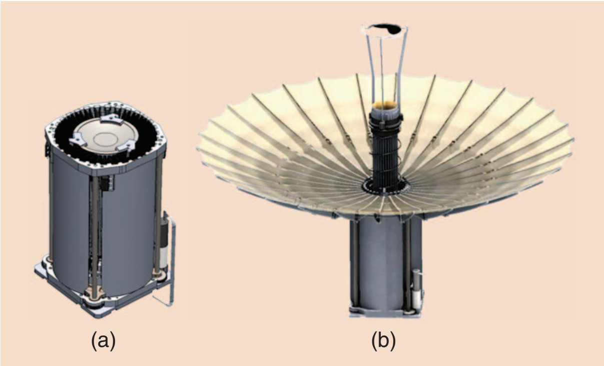


FIGURE 3. (a) The stowed antenna assembly fitting in 1.5U ($10 \times 10 \times 15$ cm^3). (b) The deployed mesh reflector antenna.

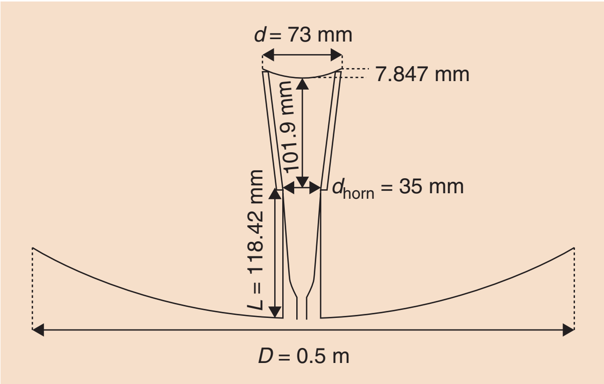


FIGURE 4. The optimized antenna design dimensions.

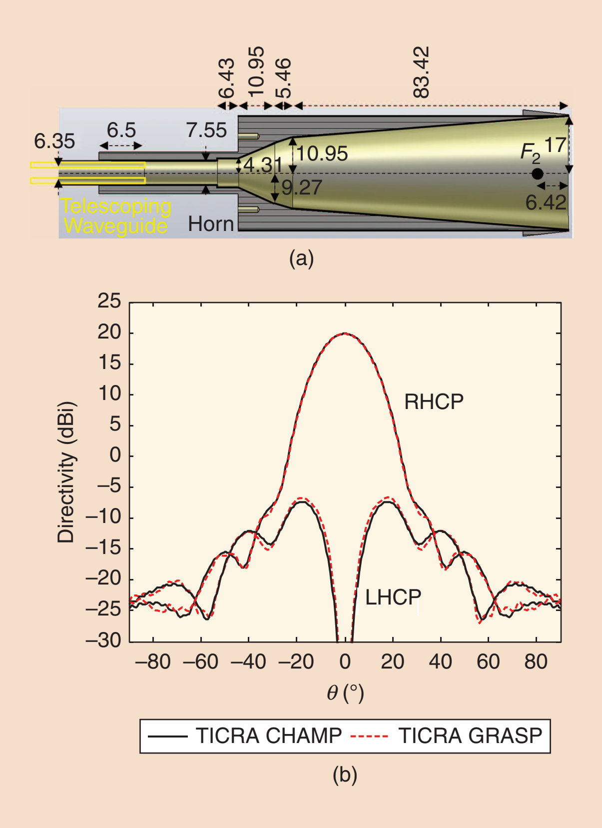


FIGURE 5. (a) The multiflare horn antenna feed design. F_2 is the second focal point of the subreflector. (b) Radiation pattern of the optimized multiflare horn feed providing a –10-dB taper at $\theta = 17°$ at 32 GHz using TICRA CHAMP (mode matching and BoR-MoM) and TICRA GRASP (MoM).

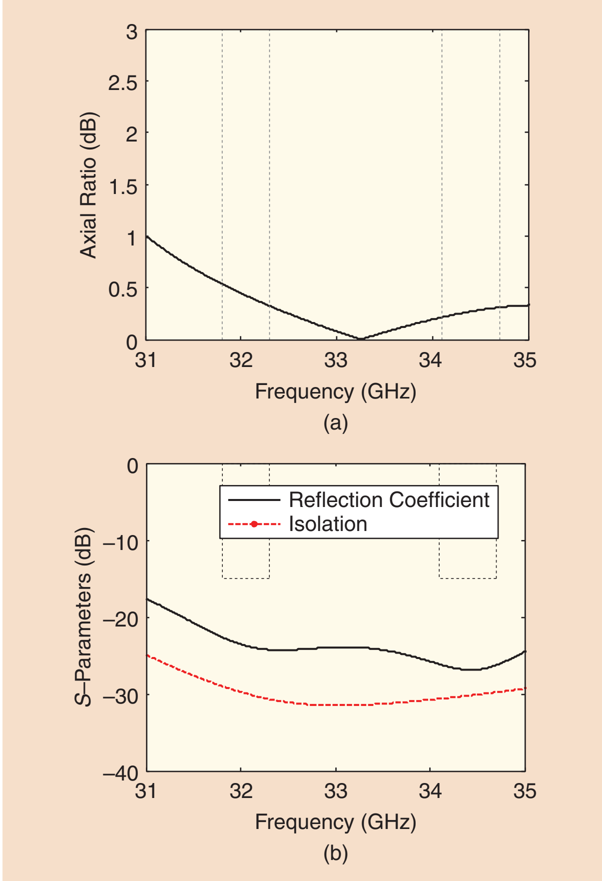


FIGURE 7. The OMT polarizer (a) axial ratio and (b) reflection coefficient of RHCP port and isolation between the two input ports.

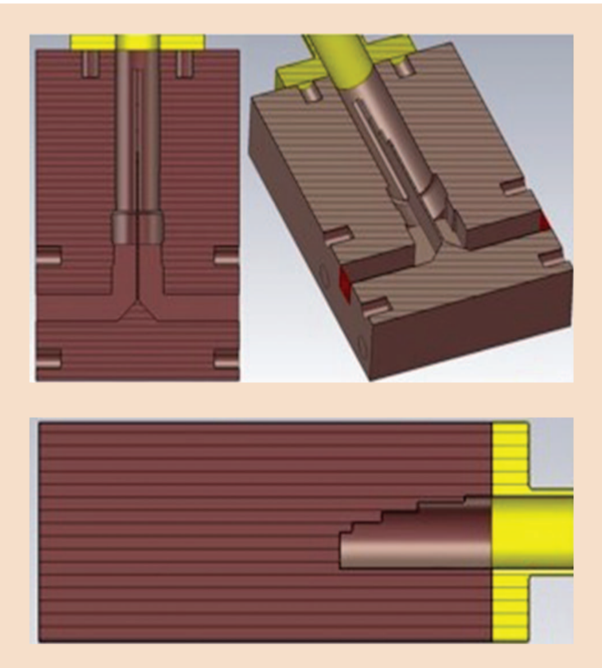


FIGURE 6. The cross sections of the OMT polarizer to excite RHCP or LHCP (only RHCP is required for DSN telecommunication).

load. Its reflection coefficient is below –20 dB and its isolation is under –30 dB at the DSN bands, as shown in Figure 7.

Because of taper and SLL, one cannot obtain an efficiency of $\eta = \eta_T \cdot \eta_S$ higher than 81% (i.e., –0.9 dB, where η_T is the taper efficiency and η_S is the spillover efficiency) [11]. Other loss contributions that should be considered and minimized include

- subreflector blockage
- strut and feed blockage
- surface aberration due to ribs
- mesh loss
- surface accuracy
- feed and waveguide loss
- mismatch loss.

The antenna performance is calculated the general reflector antenna software package's (GRASP) model shown in Figure 8. GRASP is a commercially available product from TICRA, Copenhagen, Denmark. The feed horn, combined with the three struts and the subreflector, is modeled as a method-of-moments (MoM)/multilevel fast multipole method (MLFMM) object. A waveguide port is used to excite the MoM/MLFMM object. The horn itself is defined using two piecewise linear body-of-revolution objects, with one for the interior and one for the exterior. These two objects are combined in a scatterer cluster and define the horn geometry.

TABLE 1. A GAIN TABLE AT 32 AND 34.45 GHZ AFTER COMPENSATION (30 RIBS).

	32 GHz		34.45 GHz	
	Gain (dBi)	Loss (dB)	Gain (dBi)	Loss (dB)
Ideal directivity	44.48	–	45.01	–
Spillover	43.85	0.63	44.61	0.40
Taper	43.23	0.62	43.89	0.72
Blockage	43.06	0.17	43.26	0.63
Struts	42.79	0.27	43.13	0.13
Surface mesh (40 opi)	42.54	0.25	42.88	0.25
Surface rms (±0.22 mm)	42.16	0.38	42.44	0.44
Feed loss	42.04	0.12	42.32	0.12
Feed mismatch (RL = 25 dB)	42.02	0.02	42.3	0.02
Total	42.02	2.46	42.3	2.71

RL: return loss.

The radiation pattern of the multiflare horn was simulated using TICRA Champ [MoM and body-of-revolution (BoR) MoM] and TICRA GRASP (MoM) to validate the result. Both are in excellent agreement, as can be seen in Figure 5. After calculating and validating the horn radiation pattern, a scatterer cluster including the three struts and the subreflector is created and used as a MoM/MLFMM object. Then, the current is calculated at the surface of the mesh reflector using physical optics (PO).

The loss contributions are summarized in Table 1. It was demonstrated in [2] that a surface root mean square (rms) of 0.22 mm can be achieved. Ruze's equation predicts losses of 0.38 and 0.44 dB at 32 and 34.45 GHz, respectively. This antenna uses 40 openings-per-inch (opi) mesh knitted from 0.0008-in-diameter gold-plated tungsten wire. The 0.25-dB losses resulting from the mesh opi were numerically assessed using GRASP; the measurement performed in [2] at 35.75 GHz agrees with this calculated loss value. The optimized gain is estimated to be 42.02 and 42.30 dBi at 32 and 34.45 GHz, respectively. This translates into an aperture efficiency of 57 and 54% at 32 and 34.45 GHz, respectively.

The reflection coefficient of the reflector antenna is shown in Figure 9. For telecommunication, a return loss of 25 dB is excellent, because less than 1% of the power is reflected back. The radiation pattern was measured in the near-field anechoic chamber of NASA's Jet Propulsion Laboratory. The calculated and measured radiation patterns in azimuth and elevation planes at 32 and 34.45 GHz are shown in Figures 10 and 11. The maximum SLL is –20 and –18 dB at 32 and 34.45 GHz, respectively; the calculated and measured results are in good agreement (Table 2). The measured gains are 42.0 and 42.4 dBi at 32 and 34.45 GHz, respectively. This translates into measured efficiencies of 55% and

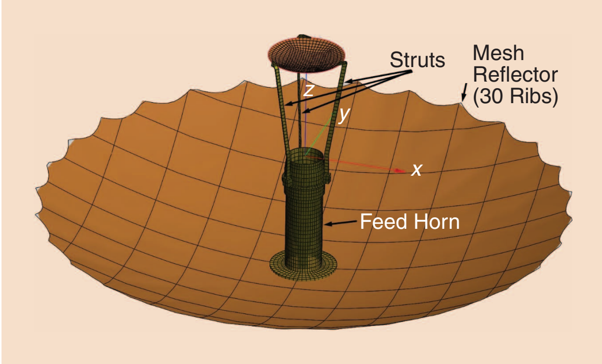


FIGURE 8. The antenna model in TICRA GRASP. The feed/struts/subreflector are an MoM object. The mesh reflector is a PO object.

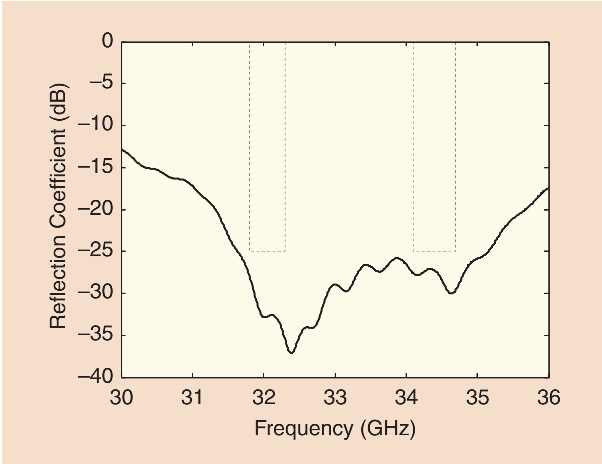


FIGURE 9. The reflection coefficient of the reflector mesh antenna including its polarizer. Solid line: simulated result; dashed line: frequency band requirement.

57% at 32 and 34.45 GHz, respectively. This is a good efficiency for this class of symmetric, Cassegrain, mesh, deployable antennas with highly demanding packaging constraints.

MECHANICAL DESIGN

MECHANICAL DESIGN CHALLENGES

The deployment mechanism has been improved significantly compared to that published in [2], and the modifications are highlighted here. The three primary challenges in designing a mesh deployable antenna are as follows:

1) creating a configuration that balances RF performance and stowed size
2) developing a method to deploy the antenna to a precise surface (with 0.22-mm rms) that can still fold in a small volume
3) applying adequate deployment force to stretch the mesh while avoiding a high-impact deployment.

The design of the proposed deployable antenna was mainly driven by the stowage volume constraining the following components:

1) height between the tip of the horn and subreflector
2) height of the horn

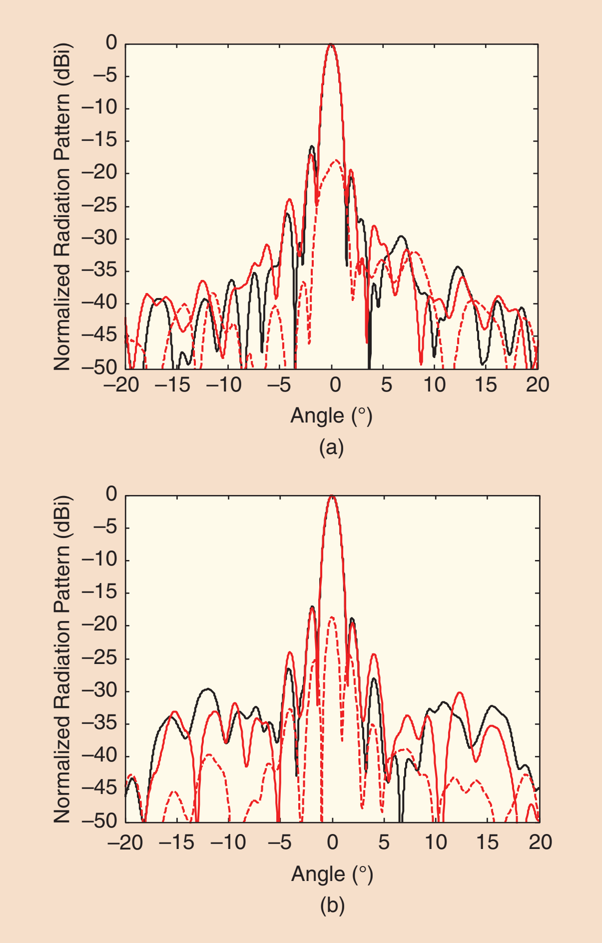


FIGURE 10. A calculated (black) and measured (red) mesh reflector antenna radiation pattern at 32 GHz. (a) $\varphi = 0°$. (b) $\varphi = 90°$. Solid line: copolarization; dashed line: cross polarization.

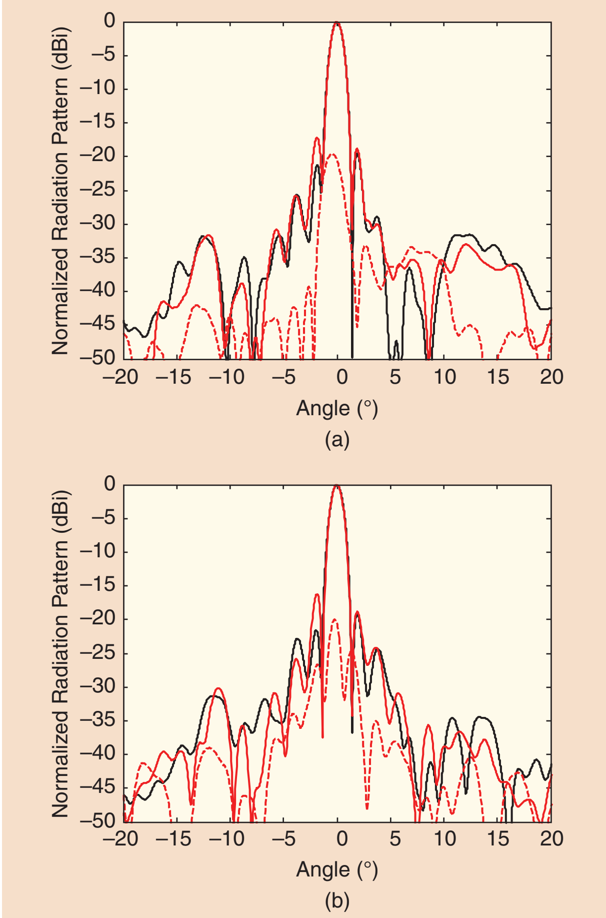


FIGURE 11. A calculated (black) and measured (red) mesh reflector antenna radiation pattern at 34.45 GHz. (a) $\varphi = 0°$. (b) $\varphi = 90°$. Solid line: copolarization; dashed line: cross polarization.

3) eccentricity of the subreflector
4) *F*/*D* ratio of the main reflector
5) number of ribs.

The next fundamental challenge after selecting the geometric architecture was to deploy the antenna with a surface accuracy of 0.22-mm rms. A surface of this accuracy would result in RF losses due to mechanical imperfections of less than 0.39 dB. To achieve this surface accuracy, the ribs were machined to a parabolic profile with an error of 0.13 mm or less. These ribs were then precision bonded into hinges on an extremely accurate bonding fixture to remove any machining errors. The rib hinges also contained a precision stop, located on the opposite side of the hinge from the hinge pin. To assemble the antenna, the middle hinges were assembled first. The root (inner) and tip (outer) ribs were then bonded to the middle hinges using the aforementioned bonding fixture. Next, the root rib hinges and bonded ribs were placed on a parabolic bonding fixture to finalize the assembly of the antenna skeleton. Finally, after the structure was completed, the mesh was stitched onto the ribs with more than 2,000 hand stitches.

TABLE 2. MEASURED AND CALCULATED GAIN AND EFFICIENCY.

	Gain (dBi)		Efficiency (%)	
	Calculations	Measurements	Calculations	Measurements
32 GHz	42.0	42.0	57	57
34.45 GHz	42.3	42.4	54	55

A key factor in determining the deployment force is the amount of force required to tension the mesh. Compared to the lighter mesh often used on L- or S-band antennas (10–30 opi), the 40-opi mesh is much denser and requires greater force to tension it on deployment. To fully stretch the mesh, calculation and experiments have shown that each rib requires 12.1 N · cm of torque at its base. When added across 30 ribs, a total preload of 290 N of force would be required to deploy.

DEPLOYMENT MECHANISM DESIGN

SmallSat antennas traditionally use strain energy of springs for deployment. If one were to compress a spring that would travel the full 16 cm of required deployment length, produce 290 N when fully deployed, and stow compactly, this spring would produce well over 1,000 N of force when stowed. This would have resulted in an extremely violent deployment and therefore necessitated the invention of a new innovative deployment mechanism for SmallSats.

The first solution concept used gas to deploy the antenna by pushing it upward in the canister, such as a piston in an engine. The gas pushed up a central hub until the ribs cleared the edge and were then forced to open by a lever arm. A prototype of the gas-powered deployment system was built and tested. Gas was pumped into the canister using an air compressor [2]. Unfortunately, near the end of the deployment test, just before the tip ribs deployed, the antenna tilted by approximately 5° because the gas actuation mechanism did not provide any features to keep the antenna perpendicular to the canister. This tilt in the deployment prevented latching of the antenna. Therefore, when pressure was lost after deployment, the antenna did not keep its desired shape or precision. Had this occurred in orbit, it would have significantly degraded performance. Furthermore, unlike the air compressor test, in which gas was pumped into the canister during the course of 1 min, it was found that available cold gas generators and sublimating powders released within one tenth of 1 s. This would likely result in a large undesirable impact on the antenna, similar to a spring deployment. Concepts for slowing the gas release with orifices and valves were considered but were eliminated because of their complexity. Therefore, a new deployment concept was needed.

It was ultimately determined that the best approach would be a motor and lead screw deployment mechanism. As the motor turns the lead screw, the base of the antenna would be slowly raised. The best place to put a single lead screw would have been in the center of the antenna, but this volume was already consumed by the feed and waveguide. To circumvent this problem, four lead screws were used, with one placed in each corner of the antenna cylinder (Figure 12). To keep the four lead screws synchronized, a single motor was used that had a unique planetary gear system. A single sun gear kept four planet gears (each attached to a lead screw) synchronized. Lead screws offer a very high gear ratio, enabling a small motor to produce the large deployment force required.

The deployment sequence began with the antenna lead screws slowly driving the antenna hub upward [Figure 12(a) and (b)]. When the antenna base reached the top of the canister, the root ribs interlocked with a feature near the top of the canister, pulling them outward [Figure 12(c)]. Then, the tip ribs deployed using a constant force spring located in the midrib hinge. The subreflector and struts began their deployment once the ribs were fully deployed. They telescoped along the horn using a compression spring [Figure 12(d)]. Tests have shown that the subreflector deploys to within 0.2 mm on the z axis and 0.1 mm on the x and y axes of its ideal position.

COMPARISON OF DEPLOYMENT MECHANISMS

The lead screw deployment mechanism augmentation in the stowed volume of the antenna increased from 1.5U to 1.6U, as the height increased to 16.2 cm. However, beyond a highly deterministic deployment, a higher number of additional advantages were observed on a motorized lead screw deployment than on a gas deployment. First, the motorized deployment eliminated the

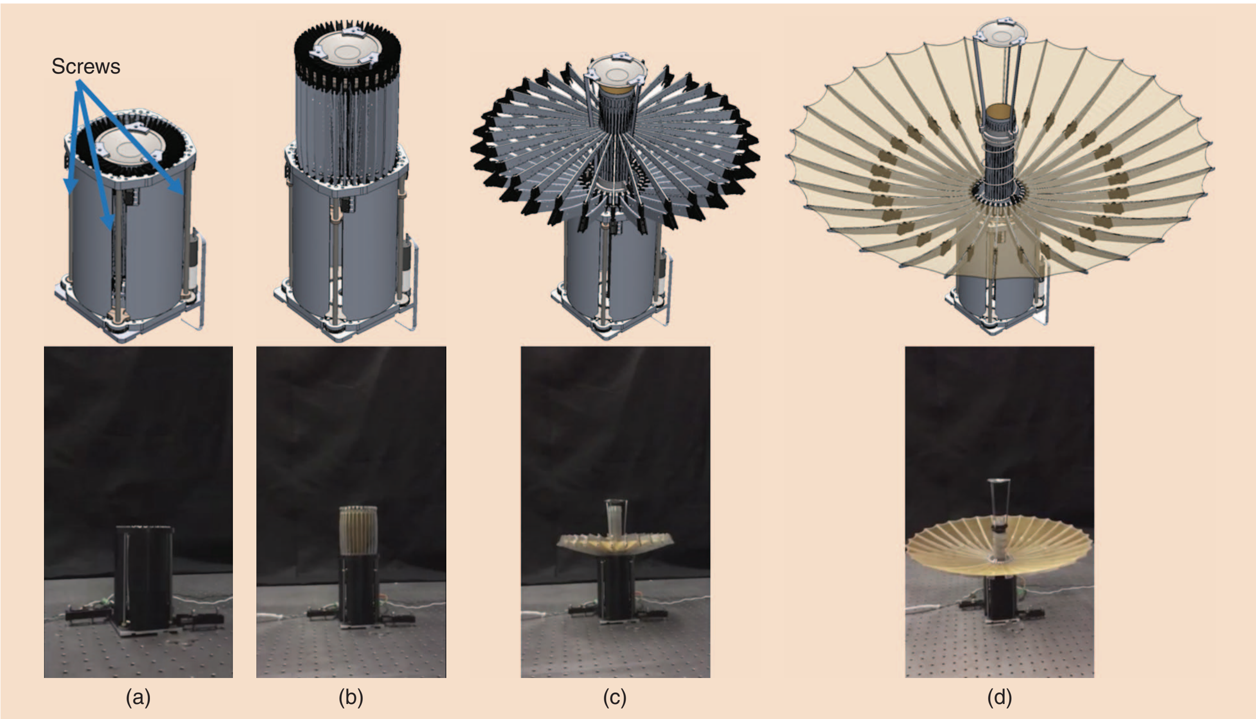


FIGURE 12. The mesh reflector antenna deployment sequence. (a) The antenna folded. (b) The antenna hub is fully pushed out of the canister. (c) The ribs are deploying. (d) The antenna fully deployed.

need for a launch lock, because the lead screws hold the antenna in place. Second, latches at the end of deployment were eliminated as once again the lead screw held the hub in place. Third, the motor now provided deployment feedback, because it was possible to determine the number of rotations needed to deploy the antenna. Fourth, there was no longer a canister of pressurized gas at the end of deployment, which, if a leak were to occur, could negatively impact the attitude control of the spacecraft. Finally, deployment tests can be done in air at a much lower cost than a pressurized deployment, because cold gas generators are costly.

SUMMARY

The mechanical design balances RF performance with stowed size through careful collaboration between mechanical and RF engineering that produced a Cassegrain secondary reflector and a 30-bifolding-rib design. Precise surface accuracy was achieved through accurate machining, precision hinge stops, and use of fixtures for assembly. Finally, adequate deployment force with a low impact was achieved by implementing a motorized lead screw design. Although some aspects of these mechanical developments have been discussed in [12], the key deployment processes were highlighted here for completeness.

CONCLUSIONS

CubeSats are strong candidates for deep-space missions and will require HGAs to enable a practical telecom system. This article presents a novel deployable HGA compatible with NASA's DSN at Ka-band frequencies. The optimized 0.5-m deployable mesh reflector antenna that evolved from our previous LEO radar antenna design fits in a limited volume of 1.5U ($10 \times 10 \times 15$ cm^3) and demonstrates excellent performance for telecommunication.

A prototype antenna was fabricated and tested. The measured gains were 42.0 and 42.4 dBi at 32 and 34.45 GHz, respectively, which is in excellent agreement with predicted values. This corresponds to measured efficiencies of 57% and 55% at 32 and 34.45 GHz, respectively. These are very satisfactory performance characteristics for this class of symmetric, Cassegrain mesh, deployable antennas with severely restricted packaging constraints operating at Ka-band with circular polarizations.

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